

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250282254

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

KODAMA; Maroi et al.

WORK VEHICLE

Abstract

A work vehicle provided with a traveling machine body in which a battery is disposed includes a first external power supply mode and a second external power supply mode in which electric power can be supplied from the battery to the outside of the traveling machine body. In the first external power supply mode, movement of the traveling machine body is prohibited, and in the second external power supply mode, movement of the traveling machine body is permitted.

Inventors: KODAMA; Maroi (Oakayama-shi, JP), SUZUKI; Shogo (Oakayama-shi, JP), ISHIDA; Tomoki (Oakayama-shi, JP)

Applicant: Yanmar Holdings Co., Ltd. (Osaka-shi, JP)

Family ID: 94687960

Assignee: Yanmar Holdings Co., Ltd. (Osaka-shi, JP)

Appl. No.: 19/072743

Filed: March 06, 2025

Foreign Application Priority Data

JP	2024-036079	Mar. 08, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: B60L55/00 (20190101); B60L50/60 (20190101); G05D1/223 (20240101); G05D105/15 (20240101); H02J1/14 (20060101)

U.S. Cl.:

CPC B60L55/00 (20190201); B60L50/60 (20190201); G05D1/223 (20240101); H02J1/14 (20130101); B60L2200/40 (20130101); B60L2260/00 (20130101); G05D2105/15

Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a work vehicle.

BACKGROUND ART

[0002] A grass mower including a traveling machine body that performs grass mowing work, and being manually operable by using a transmitter (remote control device) is known as a conventional art (see, for example, Patent Document 1).

PRIOR ART DOCUMENT

Patent Document

[0003] Patent Document 1: Japanese Unexamined Patent Application Publication No. 2019-106941

SUMMARY OF INVENTION

Technical Problem

[0004] In recent years, there has also been proposed an electric work vehicle in which a battery that stores electric power is disposed in a traveling machine body, and which is driven by using electric power stored in the battery, as an energy source. Some electric work vehicles are provided with an external power supply function that supplies electric power stored in a battery to the outside of a traveling machine body, and the traveling machine body can also be used as a (mobile) power source.

[0005] When a traveling machine body is used as a power source, it is conceivable that a user using the power source is located near the traveling machine body. However, in this case, for example, when the traveling machine body is moved due to an erroneous operation of a remote control device or the like, the traveling machine body and a person may come into contact with each other. On the other hand, if movement of the traveling machine body is restricted in a state where external power supply is possible, the user may feel inconvenience.

[0006] The present invention has been made to solve the above problems, and an object of the present invention is to provide a technique capable of securing safety and convenience of a traveling machine body in a state where external power supply is possible.

Solution to Problem

[0007] A work vehicle according to one aspect of the present invention is a work vehicle provided with a traveling machine body in which a battery is disposed. The work vehicle includes a first external power supply mode and a second external power supply mode in which electric power can be supplied from the battery to the outside of the traveling machine body. In the first external power supply mode, movement of the traveling machine body is prohibited, and in the second external power supply mode, movement of the traveling machine body is permitted.

Advantageous Effects of Invention

[0008] According to the above configuration, safety and convenience of a traveling machine body in a state where external power supply is possible can be secured.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a diagram illustrating a schematic configuration of a work vehicle according to an embodiment of the present invention.

[0010] FIG. 2 is a plan view illustrating a configuration of a remote control device of the work vehicle.

[0011] FIG. **3** is a block diagram schematically illustrating a configuration of an electrical system and a hydraulic system of the work vehicle.

[0012] FIG. **4** is a block diagram schematically illustrating a configuration of a control device provided in the work vehicle.

[0013] FIG. **5** is a diagram illustrating a configuration of a mode selection screen to be displayed on an operation unit provided in a traveling machine body of the work vehicle.

[0014] FIG. **6** is a diagram illustrating a configuration of the mode selection screen.

[0015] FIG. **7** is a diagram illustrating a configuration of a state setting screen to be displayed on the operation unit.

[0016] FIG. **8** is a diagram illustrating a configuration of the state setting screen.

[0017] FIG. **9A** is a flowchart illustrating an upstream side of a flow from start to finish of an external power supply function of the traveling machine body.

[0018] FIG. **9B** is a flowchart illustrating a downstream side of the flow from start to finish of the external power supply function.

[0019] FIG. **10** is an explanatory diagram illustrating an example of movement of the traveling machine body in a case where a movement instruction for the traveling machine body is present in a first external power supply mode of the external power supply function.

[0020] FIG. **11** is an explanatory diagram illustrating an example of movement of the traveling machine body in a case where a movement instruction for the traveling machine body is present in a second external power supply mode of the external power supply function.

[0021] FIG. **12** is an explanatory diagram illustrating power saving control to be performed in the first external power supply mode.

DESCRIPTION OF EMBODIMENTS

[0022] An embodiment of the present invention is described below with reference to the drawings.

[1. Schematic Configuration of Work Vehicle]

[0023] FIG. **1** is a diagram illustrating a schematic configuration of a work vehicle **1** according to an embodiment of the present invention. The work vehicle **1** is used, for example, to perform work such as agricultural work and construction work. The work vehicle **1** includes a vehicle body **10**, and a remote control device **200** provided separately from the vehicle body **10**.

[0024] The vehicle body **10** includes a traveling machine body **11** that travels on the ground, a work machine **12** that performs various types of work, and a hitch portion **13** that enables to connect the work machine **12** to the traveling machine body **11**. That is, the work vehicle **1** includes the traveling machine body **11**, the work machine **12**, and the hitch portion **13**.

[0025] The remote control device **200** (also referred to a remote manipulation device) allows an operator located away from the vehicle body **10** to operate the vehicle body **10** (particularly, the traveling machine body **11**). That is, the traveling machine body **11** is remote controlled by the remote control device **200**. A configuration of the remote control device **200** is described later.

[0026] Further, the work vehicle **1** is configured to be able to travel also by automatic traveling. That is, the work vehicle **1** is configured to be switchable between traveling (also referred to as manual traveling) by remote control, and automatic traveling. The automatic traveling means that, in the work vehicle **1**, at least steering is autonomously (automatically) performed along a predetermined route (also referred to as a traveling route). Automatic traveling may be configured in such a way that, for example, at least one of adjustment of a traveling speed, and work by the work machine **12** is autonomously performed, in addition to steering. Note that, a configuration of the work vehicle **1** is not limited to the above, and, for example, traveling by remote control is enabled, and automatic traveling may be omitted.

[0027] Herein, a direction (particularly, a direction related to the vehicle body **10**) used in explanation is defined as follows. One side in one direction when the traveling machine body **11** travels straight along the one direction is assumed to be “front”, and the other side thereof is assumed to be “rear”. A side where the hitch portion **13** is not disposed in a direction along which

the traveling machine body **11** travels straight along the direction is assumed to be “front”, and the opposite side (a side where the hitch portion **13** is disposed) thereof is assumed to be “rear”. Further, a left side from the rear side toward the front side is assumed to be “left”, and a right side is assumed to be “right”. Furthermore, a gravitational direction perpendicular to the front-rear direction and the left-right direction is assumed to be an up-down direction, an upstream side in the gravitational direction is assumed to be “up”, and a downstream side is assumed to be “down”. In the drawings, the front is indicated by a symbol “F”, the rear is indicated by a symbol “B”, the right is indicated by a symbol “R”, the left is indicated by a symbol “L”, the up is indicated by a symbol “U”, and the down is indicated by a symbol “D”, as necessary. Note that, these directions are simply names to be used for explanation, and are not intended to limit an actual positional relationship and direction.

[0028] The traveling machine body **11** includes a machine body unit **111**, and a traveling unit **112** disposed below the machine body unit **111**.

[0029] The machine body unit **111** includes an outer cover **111a**, a traveling drive device **111b**, and a work machine drive device **111c**.

[0030] The traveling drive device **111b** is disposed on a front side of an inner portion covered by the outer cover **111a**. The traveling drive device **111b** includes a traveling actuator **111b1** as a drive source, and a traveling power transmission unit (not illustrated) that transmits power from the traveling actuator **111b1** to the traveling unit **112**. In the present embodiment, the traveling actuator **111b1** is an electric motor. Note that, the drive source included in the traveling drive device **111b** may be a drive source other than an electric motor, and may be, for example, an engine.

[0031] The work machine drive device **111c** is disposed on a rear side of the inner portion covered by the outer cover **111a**. The work machine drive device **111c** includes a drive actuator **111c1** as a drive source, and a power take off (PTO) power transmission unit (not illustrated) that enables to transmit power from the drive actuator **111c1** to the work machine **12**. That is, the traveling machine body **11** includes the drive actuator **111c1**. In the present embodiment, the drive actuator **111c1** is an electric motor. Note that, the drive source included in the work machine drive device **111c** may be a drive source other than an electric motor, and may be, for example, an engine. Further, in the present embodiment, an electric motor (drive source) is provided individually for the traveling drive device **111b**, and the work machine drive device **111c**. However, an electric motor (drive source) may be shared.

[0032] A battery **111d** is accommodated inside the outer cover **111a**, in addition to the traveling drive device **111b** and the work machine drive device **111c**. That is, the battery **111d** is disposed in the traveling machine body **11**. The battery **111d** is constituted of, for example, a lithium-ion battery. The battery **111d** may be constituted as a unit of a plurality of battery cells, or may be constituted of a single battery cell.

[0033] The battery **111d** stores electric power for driving the traveling actuator **111b1**, the drive actuator **111c1**, and the like. Further, in the present embodiment, electric power stored in the battery **111d** can be supplied to the outside of the traveling machine body **11**. Herein, supplying electric power from the battery **111d** to the outside of the traveling machine body **11** is referred to as “external power supply”. That is, the traveling machine body **11** in the present embodiment is provided with an external power supply function that enables external power supply. Note that, in the external power supply function, two modes (a first external power supply mode, and a second external power supply mode) are provided. That is, the traveling machine body **11** includes the first external power supply mode, and the second external power supply mode. Details of the external power supply function (the first external power supply mode, and the second external power supply mode) are described later.

[0034] As an example, a light **111e**, a positioning antenna **111f**, an alarm lamp **111g** and the like are disposed outside the outer cover **111a**.

[0035] The traveling unit **112** supports the machine body unit **111** to be able to travel. Specifically,

the traveling unit **112** includes a pair of left and right crawlers **112a**. Each of the left and right crawlers **112a** includes a truck frame **112b** extending in the front-rear direction. Each truck frame **112b** is mounted on a lower surface of the machine body unit **111**. A drive sprocket **112c** is disposed as a drive wheel at a front end of the truck frame **112b**. Power is transmitted from the traveling actuator **111b1** to the drive sprocket **112c** via the traveling power transmission unit included in the traveling drive device **111b**. A driven sprocket **112d** is disposed as a driven wheel at a rear end of the truck frame **112b**. The driven sprocket **112d** is supported to be pivotable on the truck frame **112b**. In the truck frame **112b**, a plurality of wheels **112e** are supported to be pivotable between the drive sprocket **112c** and the driven sprocket **112d**. A crawler belt **112f** is wound around the drive sprocket **112c**, the driven sprocket **112d**, and the plurality of wheels **112e**. Thus, the crawler **112a** is configured.

[0036] Each of the left and right crawlers **112a** is driven by the individual traveling actuator **111b1** included in the traveling drive device **111b**. When the pair of left and right crawlers **112a** are simultaneously driven in the same direction, the traveling unit **112** travels straight forward or backward. Whether the traveling unit **112** travels forward or backward is determined by a rotating direction of an output shaft of the traveling actuator **111b1**. The traveling unit **112** turns left or turns right, for example, by independently driving the pair of left and right crawlers **112a**.

[0037] Note that, in the present embodiment, the crawler **112a** has a configuration in which one drive wheel (drive sprocket **112c**), and one driven wheel (driven sprocket **112d**) are arranged side by side in the front-back direction, and the crawler belt **112f** is wound around the drive wheel and the driven wheel. However, another configuration may be adopted. For example, a crawler may be of a type in which a crawler belt is wound around one drive wheel and two driven wheels in a triangular shape. Further, in the present embodiment, the traveling unit **112** is of a crawler type, however, the traveling unit **112** may be of a type other than a crawler type, and, for example, may be of a wheel type.

[0038] The work machine **12** is disposed behind the hitch portion **13**. The hitch portion **13** is configured to include the work machine drive device **111c**. The work machine **12** is mounted to be able to be raised and lowered with respect to the traveling machine body **11** via the hitch portion **13**. That is, the work machine **12** is connected to the traveling machine body **11** via the hitch portion **13**. Note that, in the present embodiment, the work machine **12** is disposed behind the traveling machine body **11**. However, the present invention is not limited thereto, and, for example, the work machine **12** may be disposed in front of the traveling machine body **11**.

[0039] Further, the work machine **12** is mounted to be detachable with respect to the hitch portion **13**. That is, the work machine **12** is replaceable with various types of machines. In FIG. **1**, the work machine **12** is a tiller. The work machine **12** may be, other than a tiller, for example, a plow, a ridge making machine, a fertilizer applicator, a pesticide spraying machine, a harvester, a mowing machine (also referred to as a grass mowing machine), a snow removing machine, and the like.

[0040] Further, the work machine **12** is driven by power to be output from the work machine drive device **111c**. More specifically, the work machine **12** is driven by power of the drive actuator **111c1** to be transmitted via the PTO power transmission unit. That is, the drive actuator **111c1** drives the work machine **12**.

[0041] Note that, in the present embodiment, the work machine **12** is connected to a rear portion of the traveling machine body **11**. However, the present invention is also applicable to a work vehicle **1** in which the work machine **12** is connected to a front portion of the traveling machine body **11**. Further, the present invention is also applicable to a work vehicle **1** in which the work machine **12** is detached (from the hitch portion **13**).

[2. Configuration of Remote Control Device]

[0042] A configuration of the remote control device **200** is described with reference to FIG. **2**. FIG. **2** is a plan view illustrating a configuration of the remote control device **200**. The remote control device **200** includes a housing **201**, a power switch **202**, an antenna **203**, an operation lever **204**, an

operation switch **205**, an operation knob **206**, and a display unit **207**.

[0043] The housing **201** constitutes a main body portion of the remote control device **200**. Each of the power switch **202**, the antenna **203**, the operation lever **204**, the operation switch **205**, the operation knob **206**, and the display unit **207** described above is mounted at an appropriate position of the housing **201**. Note that, a layout illustrated in FIG. 2 is merely an example, and may be changed as appropriate.

[0044] The power switch **202** is provided at a middle portion on a front surface of the housing **201**, and enables on and off of power supply of the remote control device **200**. The power switch **202** is, for example, a seesaw switch. Note that, a power source of the remote control device **200** is, for example, a battery, a dry cell battery, or the like disposed inside the housing **201**.

[0045] The antenna **203** is provided in such a way as to protrude from a side surface (an upper side surface in FIG. 2) of the housing **201**, and enables wireless communication with the traveling machine body **11**. In a case where power supply of the remote control device **200** is in an on-state by the power switch **202**, the remote control device **200** can be wirelessly communicated with the traveling machine body **11**. In a case where power supply of the remote control device **200** is in an off-state by the power switch **202**, the remote control device **200** cannot communicate with the traveling machine body **11**. In the present embodiment, when the traveling machine body **11** cannot communicate with the remote control device **200**, the traveling machine body **11** automatically stops traveling. That is, the power switch **202** has a function as an emergency stop switch of the traveling machine body **11**. Note that, an emergency stop switch may be provided separately from the power switch **202**.

[0046] The operation lever **204** enables a traveling operation of the traveling machine body **11**, and an operation of the work machine **12**. The operation lever **204** includes a first operation lever **204a** and a second operation lever **204b** disposed laterally side by side with the power switch **202** interposed therebetween. The first operation lever **204a** (a left lever illustrated in FIG. 2) can be tilted at least in two directions (F1-B1 direction and L1-R1 direction) orthogonal to each other, which are indicated by dashed arrows in FIG. 2. Further, the first operation lever **204a** can also be tilted at least in one direction of the L1-R1 direction, while being tilted at least in one of the F1-B1 direction. Note that, the second operation lever **204b** (a right lever illustrated in FIG. 2) can also be tilted in the same direction as the first operation lever **204a**.

[0047] When the first operation lever **204a** is tilted in the F1 direction, the traveling machine body **11** can be moved forward. When the first operation lever **204a** is tilted in the B1 direction, the traveling machine body **11** can be moved backward. When the first operation lever **204a** is tilted in the L1 direction, the traveling machine body **11** can turn left. When the first operation lever **204a** is tilted in the R1 direction, the traveling machine body **11** can turn right.

[0048] When the second operation lever **204b** is tilted in the F1 direction, the work machine **12** can be raised. When the second operation lever **204b** is tilted in the B1 direction, the work machine **12** can be lowered. When the second operation lever **204b** is tilted in the L1 direction, the traveling machine body **11** can turn left. When the second operation lever **204b** is tilted in the R1 direction, the traveling machine body **11** can turn right.

[0049] Note that, a turning radius (turning amount) in a case where the operation lever is operated by the same operation amount (operation amount from a neutral position) is different between turning (left turn and right turn) by the first operation lever **204a**, and turning (left turn and right turn) by the second operation lever **204b**. More specifically, in a case where the first operation lever **204a** and the second operation lever **204b** are tilted by the same amount from a neutral position, the traveling machine body **11** can turn with a large turning radius when operated by the second operation lever **204b**, as compared with a case where the traveling machine body **11** is operated by the first operation lever **204a**. That is, the traveling machine body **11** can be gently turned by the second operation lever **204b**. Note that, a configuration of the first operation lever **204a** and the second operation lever **204b** is not limited to the above. For example, it may be configured in such

a way that the traveling machine body **11** is gently turned by the first operation lever **204a**, as compared with a case of using the second operation lever **204b**.

[0050] The operation switch **205** includes a first operation switch **205a**, and a second operation switch **205b**. The first operation switch **205a** is disposed on a side surface (an upper right side surface in FIG. 2) of the housing **201**, and enables a plurality of types of settings related to automatic traveling. The plurality of types of settings include, for example, start setting and finish setting on automatic traveling. The first operation switch **205a** is, for example, a momentary switch that can be pressed down.

[0051] The second operation switch **205b** is disposed on a side surface (an upper left side surface in FIG. 2) of the housing **201**, and enables on and off of driving of the work machine **12**. That is, when the second operation switch **205b** is operated, on and off of driving of the drive actuator **111c** is switched. The second operation switch **205b** is, for example, a toggle switch.

[0052] The operation knob **206** is disposed on a front surface (an upper side of a front surface in FIG. 2) of the housing **201**, and enables adjustment of a maximum speed of the traveling machine body **11**. More specifically, two operation knobs **206** are provided. One of the two operation knobs **206** enables adjustment of a maximum speed of the traveling machine body **11** at a time of straight traveling. The other of the two operation knobs **206** enables adjustment of a maximum speed of the traveling machine body **11** at a time of traveling while turning.

[0053] The display unit **207** is disposed on a front surface (a lower side of the front surface in FIG. 2) of the housing **201**, and displays various pieces of information to be notified to an operator. The various pieces of information include, for example, indication of a traveling speed of the traveling machine body **11**, and the like. The display unit **207** is, for example, a liquid crystal display device, an organic EL display device, and the like.

[3. Configuration of Electrical System and Hydraulic System of Work Vehicle]

[0054] A configuration of an electrical system and a hydraulic system of the work vehicle **1** is described with reference to FIG. 3. FIG. 3 is a block diagram schematically illustrating a configuration of an electrical system and a hydraulic system of the work vehicle **1**. Note that, in FIG. 3, a solid line indicates an electric power supply path, a dashed line indicates a hydraulic oil supply path, and one-dot chain line indicates a signal path.

[0055] The work vehicle **1** includes a plurality of pieces of electric equipment **113**, and a plurality of pieces of hydraulic equipment **114**. The plurality of pieces of electric equipment **113** include a control device **113a**, a charger **113b**, a first inverter **113c**, a relay box **113d**, a junction box **113e**, a second inverter **113f**, an external power supply port **113g**, a DC-DC converter **113h**, a low voltage battery **113j**, a positioning communication unit **113k**, a communication processing unit **113m**, and an operation unit **113n**. The plurality of pieces of electric equipment **113** are disposed in the machine body unit **111** (see FIG. 1) of the traveling machine body **11**.

[0056] The plurality of pieces of electric equipment **113** receive electric power supply from the battery **111d**. More specifically, each piece of equipment included in the plurality of pieces of electric equipment **113** is directly connected to the battery **111d**, or electrically connected to the battery **111d** via another piece of equipment included in the plurality of pieces of electric equipment **113**. For example, the relay box **113d** included in the plurality of pieces of electric equipment **113** is directly connected to the battery **111d**. Further, the first inverter **113c** included in the plurality of pieces of electric equipment **113** is electrically connected to the battery **111d** via the relay box **113d**, and the junction box **113e**.

[0057] The control device **113a** is a computer device that performs electrical control of each unit of the work vehicle **1**. A configuration of the control device **113a** is described later. The charger **113b** converts alternate-current voltage to be supplied from an external power source (not illustrated) via a charging cable (not illustrated) into direct-current voltage. Note that, a charging port (not illustrated) for connecting the charging cable is provided in the machine body unit **111** of the traveling machine body **11**.

[0058] The first inverter **113c** converts direct-current voltage to be supplied from the battery **111d** into alternate-current voltage, and supplies the alternate-current voltage to the traveling actuator **111b1**, and the drive actuator **111c1**. Thus, the traveling actuator **111b1** and the drive actuator **111c1** are driven. Supply of alternate-current voltage from the first inverter **113c** to the traveling actuator **111b1**, and the drive actuator **111c1** is performed based on a rotation command to be output from the control device **113a**.

[0059] The relay box **113d** is a battery control unit that controls input/output of the battery **111d** by controlling an internal battery relay. The junction box **113e** is configured to include a charger relay, an inverter relay, a fuse, and the like. Voltage to be output from the charger **113b** is supplied to the battery **111d** via the junction box **113e**, and the relay box **113d**. Thus, electric power is stored in the battery **111d**. Further, voltage to be output from the battery **111d** is supplied to the first inverter **113c** via the relay box **113d**, and the junction box **113e**. In addition to the above, voltage to be output from the battery **111d** is also supplied to the second inverter **113f** via the relay box **113d**, and the junction box **113e**.

[0060] The second inverter **113f** converts direct-current voltage (e.g., 350V) to be supplied from the battery **111d** into alternate-current voltage (e.g., 100 V), and supplies the alternate-current voltage to the external power supply port **113g**. Supply of alternate-current voltage from the second inverter **113f** to the external power supply port **113g** is performed based on a command to be output from the control device **113a**.

[0061] The external power supply port **113g** is configured to be connectable to an external connector (not illustrated) of the traveling machine body **11**, and supplies electric power to the outside of the traveling machine body **11**. That is, for example, connecting the external power supply port **113g**, and external equipment (not illustrated) of the traveling machine body **11** to each other by using a power supply cable (not illustrated) including the connector enables to supply electric power of the battery **111d** to the equipment. In other words, external power supply is enabled. The equipment includes, for example, an electric work machine (a pesticide spraying machine, a harvester, and the like), a mobile terminal (a personal computer, a smartphone, and the like). Therefore, in the present embodiment, a state in which electric power is supplied to the external power supply port **113g** is referred to as a “state in which external power supply is possible”. The external power supply port **113g** is, for example, provided in a right side portion of the machine body unit **111**.

[0062] Note that, the connector may be provided in the equipment itself, and, in this case, electric power is supplied to the equipment by directly connecting the equipment to the external power supply port **113g**.

[0063] Further, in the present embodiment, the first inverter **113c**, and the second inverter **113f** are individually provided. However, the present invention is not limited to this configuration. For example, one inverter may be provided for the traveling actuator **111b1**, the drive actuator **111c1**, and the external power supply port **113g**.

[0064] The DC-DC converter **113h** (also simply referred to as a converter) lowers direct-current high voltage (e.g., 350 V) to be supplied from the battery **111d** via the junction box **113e** to low voltage (e.g., 12 V). The low voltage battery **113j** is, for example, constituted of a lead battery. The low voltage battery **113j** is connected to the DC-DC converter **113h**, and stores electric power in which voltage is lowered by the DC-DC converter **113h**.

[0065] Electric power whose voltage is lowered by the DC-DC converter **113h** is supplied to each of the positioning communication unit **113k**, the communication processing unit **113m**, and the operation unit **113n**, and each of the positioning communication unit **113k**, the communication processing unit **113m**, and the operation unit **113n** is actuated. Note that, a configuration of each of the positioning communication unit **113k**, the communication processing unit **113m**, and the operation unit **113n** is described later.

[0066] The plurality of pieces of hydraulic equipment **114** are disposed in the traveling machine

body **11**, particularly, on a lower side of a rear portion of the machine body unit **111**. The plurality of pieces of hydraulic equipment **114** include a hydraulic pump **114a**, a direction switching valve **114b**, and a lifting actuator **114c**. That is, the traveling machine body **11** includes the hydraulic pump **114a**, the direction switching valve **114b**, and the lifting actuator **114c**.

[0067] In the present embodiment, the hydraulic pump **114a** internally includes an electric motor (not illustrated), and is constituted of an electric hydraulic pump to be driven by the electric motor. More specifically, the hydraulic pump **114a** is connected to the DC-DC converter **113h**, and driven by electric power to be output from the DC-DC converter **113h**. Note that, the hydraulic pump **114a** is not limited to an electric type, and, for example, may be a mechanical type. That is, it may be configured in such a way that an engine provided separately from the hydraulic pump **114a** serves as a drive source, an output shaft of the engine and an input shaft of the hydraulic pump **114a** are connected to each other, and the hydraulic pump **114a** is driven by power to be output from the engine.

[0068] Further, the hydraulic pump **114a** is connected to a hydraulic oil tank (not illustrated) that stores hydraulic oil. When electric power is supplied from the DC-DC converter **113h**, and the hydraulic pump **114a** is driven, hydraulic oil in the hydraulic oil tank is supplied to the lifting actuator **114c** via the direction switching valve **114b**. The direction switching valve **114b** controls a flow direction and a flow rate of hydraulic oil to be supplied to the lifting actuator **114c**.

[0069] In the present embodiment, the lifting actuator **114c** is constituted of a hydraulic cylinder. More specifically, a base end portion of the lifting actuator **114c** is connected to the machine body unit **111**, and a lead end portion thereof is connected to the hitch portion **13** (see FIG. 1). When the lifting actuator **114c** is expanded and contracted, the hitch portion **13** pivotally rotates in the up-down direction with respect to the traveling machine body **11**. The work machine **12** is mounted to the hitch portion **13** as described above. Therefore, expanding and contracting the lifting actuator **114c** enables to move the work machine **12** in the up-down direction. That is, the lifting actuator **114c** raises and lowers the work machine **12**. Note that, a configuration of the lifting actuator **114c** is not limited to that of a hydraulic cylinder, and, for example, the lifting actuator **114c** may be an electric cylinder.

[4. Configuration of Control Device]

[0070] A configuration of the control device **113a** is described with reference to FIG. 4. FIG. 4 is a block diagram schematically illustrating a configuration of the control device **113a**. Note that, in FIG. 4, constituent elements necessary to describe the features of the present embodiment are illustrated, and description on general constituent elements is omitted.

[0071] The control device **113a** is configured to include, for example, an arithmetic device, an input/output unit, and a storage unit **113a1**. The arithmetic device is, for example, a processor or a microprocessor. The storage unit **113a1** is a main storage device as exemplified by a read only memory (ROM) or a random access memory (RAM). The storage unit **113a1** may further include an auxiliary storage device as exemplified by a hard disk drive (HDD) or a solid state drive (SSD). Various types of programs, data, and the like are stored in the storage unit **113a1**. The arithmetic device reads the various type of programs from the storage unit **113a1**, and performs arithmetic processing in accordance with a program. The program stored in the storage unit **113a1** may be provided, for example, by a computer-readable non-volatile recording medium. As another example, the program may be provided from a program providing server via a communication line such as the Internet.

[0072] By cooperation of the hardware and software, the control device **113a** can act as a traveling control unit **113a2**, a work machine control unit **113a3**, and an external power supply control unit **113a4**. The control device **113a** may be configured of one piece of hardware, or may be configured of a plurality of pieces of hardware communicable to one another.

[0073] Note that, as described above, each of the functional units **113a2** to **113a4** included in the control device **113a** may be achieved by causing the arithmetic device to perform arithmetic

processing in accordance with a program, namely, by software, however, may be achieved by another method. At least one of the functional units **113a2** to **113a4** may be achieved by using, for example, an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or the like. That is, at least one of the functional units **113a2** to **113a4** may be achieved by hardware with use of a dedicated IC or the like. Further, at least one of the functional units **113a2** to **113a4** may be achieved by combining software and hardware. Further, each of the functional units **113a2** to **113a4** is a conceptual configuration. Therefore, a function to be executed by one constituent element may be distributed among a plurality of constituent elements, or functions included in a plurality of constituent elements may be integrated in one constituent element.

[0074] The traveling control unit **113a2** controls the traveling drive device **111b** (see FIG. 1) depending on whether the work vehicle **1** is traveling manually or traveling automatically (via the first inverter **113c**). More specifically, during manual traveling, the traveling control unit **113a2** controls the traveling drive device **111b** (particularly, the traveling actuator **111b1**), based on a traveling instruction (e.g., a forward traveling instruction by the first operation lever **204a**) from the remote control device **200**. During automatic traveling, the traveling control unit **113a2** automatically (autonomously) controls the traveling drive device **111b** (particularly, the traveling actuator **111b1**) in such a way that the traveling machine body **11** travels along a predetermined route.

[0075] The work machine control unit **113a3** controls the hydraulic equipment **114** (particularly, the hydraulic pump **114a** and the direction switching valve **114b**), based on a lifting instruction from the remote control device **200** (see FIG. 3). The lifting instruction is achieved, for example, by allowing an operator to tilt the second operation lever **204b** in the F1 direction or in the B1 direction (see FIG. 2).

[0076] More specifically, when an operator tilts the second operation lever **204b** in the F1 direction, electric power is supplied from the DC-DC converter **113h** to the hydraulic pump **114a**, and the hydraulic pump **114a** is driven. Note that, in a case where the second operation lever **204b** is at a neutral position in the F1-B1 direction (in a case where the second operation lever **204b** is not operated in the F1 direction or in the B1 direction), supply of electric power from the DC-DC converter **113h** to the hydraulic pump **114a** is cut off. Further, the direction switching valve **114b** switches a flow direction of hydraulic oil to be supplied to the lifting actuator **114c** in such a way that the lifting actuator **114c** is expanded. Thus, the lifting actuator **114c** is expanded, and the work machine **12** is raised.

[0077] Likewise, when the operator tilts the second operation lever **204b** in the B1 direction, the hydraulic pump **114a** is driven, and the direction switching valve **114b** switches a flow direction of hydraulic oil in such a way that the lifting actuator **114c** is contracted. Thus, the lifting actuator **114c** is contracted, and the work machine **12** is lowered.

[0078] Further, the work machine control unit **113a3** controls the work machine drive device **111c** (via the first inverter **113c**), based on an on-off instruction (e.g., an on instruction by the second operation switch **205b**) from the remote control device **200** (see FIG. 1). In particular, the work machine control unit **113a3** controls the drive actuator **111c1**.

[0079] The external power supply control unit **113a4** performs control related to the external power supply function. For example, the external power supply control unit **113a4** controls the second inverter **113f**, based on a command from the operation unit **113n**, which is described later.

[0080] The control device **113a** is connected to the positioning communication unit **113k**, the communication processing unit **113m**, and the operation unit **113n**. Further, the control device **113a** is connected to a sensor **113p**, in addition to these components. Note that, the sensor **113p** is also included in the electric equipment **113**, as well as the positioning communication unit **113k** and the like.

[0081] The sensor **113p** detects information related to the vehicle body **10**, and outputs the detected information to the control device **113a**. In the present embodiment, the sensor **113p** includes a

plurality of types of sensors. Each of the plurality of types of sensors is connected in such a way that a signal can be input to the control device **113a**. The plurality of types of sensors include, for example, an inertial measurement device, an obstacle sensor, a vehicle speed sensor, a lifting position sensor, and the like.

[0082] The inertial measurement device is a device including a 3-axis angular velocity sensor, and a 3-direction acceleration sensor, and capable of measuring a pose of the traveling machine body **11**. The obstacle sensor is a sensor that detects an obstacle present in the vicinity of the vehicle body **10**, and may be, for example, an ultrasonic sensor, a camera, a radar, light detection and ranging (LiDAR), or the like. The vehicle speed sensor is a sensor that detects a vehicle speed of the traveling machine body **11**. The lifting position sensor is a sensor that detects a lifting position (height from the ground) of the work machine **12**.

[0083] The positioning communication unit **113k** includes the positioning antenna **111f** (see FIG. 1), and acquires a position of the vehicle body **10**, for example, as information on a latitude and a longitude with use of a positioning signal received from a positioning satellite by the positioning antenna **111f**. The positioning communication unit **113k** performs positioning, for example, by using a known real time kinematic GNSS (RTK-GNSS) method, after receiving a positioning signal from an unillustrated reference station by an appropriate method. The positioning communication unit **113k** outputs position information on the vehicle body **10** to the control device **113a**. Note that, the positioning communication unit **113k** may perform positioning, for example, by using another method such as a differential GNSS (DGNSS) method. Further, the work vehicle **1** may be configured to include, for example, a quantum compass capable of positioning, in place of or in addition to the positioning communication unit **113k**.

[0084] Note that, the positioning communication unit **113k**, and some of the sensors **113p** (e.g., the inertial measurement device) are used in a case where the work vehicle **1** automatically travels. That is, the positioning communication unit **113k**, and some of the sensors **113p** are not essential for the work vehicle **1** to travel based on an instruction from the remote control device **200**.

[0085] The communication processing unit **113m** performs communication with the remote control device **200** via the communication antenna **113m1**. The communication antenna **113m1** is an antenna for performing wireless communication with the remote control device **200**. For wireless communication, a wireless local area network (LAN) such as Wi-Fi (registered trademark) may be used.

[0086] The operation unit **113n** is constituted of a touch panel. The touch panel allows the operation unit **113n** to perform various settings on the work vehicle **1** (particularly, the traveling machine body **11**), and display various pieces of information on the work vehicle **1** (particularly, the traveling machine body **11**). That is, a screen (image) for performing the various settings is displayed on the operation unit **113n**. The various settings include settings on the external power supply function. Note that, a display unit other than the operation unit **113n** may be provided to display various pieces of information on the work vehicle **1**. Further, the operation unit **113n** is not limited to the above configuration, and, for example, may be constituted of a switch, a dial, a lever, or the like. Hereinafter, a configuration of a setting screen for the external power supply function to be displayed on the operation unit **113n** is described.

[5. Configuration of Setting Screen for External Power Supply Function]

[0087] A setting screen for the external power supply function includes a mode selection screen **113n1** for performing selection of an external power supply mode (a first external power supply mode, and a second external power supply mode), and a state setting screen **113n2** for performing state settings on the external power supply function. First, a configuration of the mode selection screen **113n1** is described with reference to FIGS. 5 and 6. FIGS. 5 and 6 are diagrams illustrating a configuration of the mode selection screen **113n1**.

[0088] The mode selection screen **113n1** includes a first external power supply mode selection icon **IC1**, and a second external power supply mode selection icon **IC2**. Further, the first external power

supply mode selection icon IC1 includes a first external power supply mode display icon IC1a, and the second external power supply mode selection icon IC2 includes a second external power supply mode display icon IC2a. Note that, in FIGS. 5 and 6, in the first external power supply mode display icon IC1a and the second external power supply mode display icon IC2a, white filling (no filling) indicates light extinction, and black filling indicates lighting (or blinking).

[0089] The first external power supply mode selection icon IC1 enables to select the first external power supply mode in the external power supply function. The first external power supply mode display icon IC1a is turned on (or blinked), when the first external power supply mode is selected in the external power supply function (particularly, see FIG. 5). On the other hand, the first external power supply mode display icon IC1a is turned off, when the first external power supply mode is not selected, that is, the second external power supply mode is selected in the external power supply function (particularly, see FIG. 6).

[0090] The second external power supply mode selection icon IC2 enables to select the second external power supply mode in the external power supply function. The second external power supply mode display icon IC2a is turned on (or blinked), when the second external power supply mode is selected in the external power supply function (particularly, see FIG. 6). On the other hand, the second external power supply mode display icon IC2a is turned off, when the second external power supply mode is not selected, that is, when the first external power supply mode is selected in the external power supply function (particularly, see FIG. 5).

[0091] Note that, in the present embodiment, the external power supply function is maintained in a disabled state (not enabled) alone by an operation of the first external power supply mode selection icon IC1 or the second external power supply mode selection icon IC2. This is described in detail later.

[0092] Next, a configuration of the state setting screen 113n2 is described with reference to FIGS. 7 and 8. FIGS. 7 and 8 are diagrams illustrating a configuration of the state setting screen 113n2. The state setting screen 113n2 includes an enabled state setting icon IC3, and a disabled state setting icon IC4. Further, the enabled state setting icon IC3 includes an enabled state display icon IC3a, and the disabled state setting icon IC4 includes a disabled state display icon IC4a. Note that, in FIGS. 7 and 8, similarly to FIGS. 5 and 6, in the enabled state display icon IC3a and the disabled state display icon IC4a, white filling (no filling) indicates light extinction, and black filling indicates lighting (or blinking).

[0093] The enabled state setting icon IC3 enables setting on an enabled state (start) of the external power supply function. The enabled state display icon IC3a is turned on (or blinked), when the external power supply function is in an enabled state (in a state where external power supply is possible) (particularly, see FIG. 7), and is turned off, when the external power supply is in a disabled state (in a state where external power supply is not possible) (particularly, see FIG. 8).

[0094] The disabled state setting icon IC4 enables setting on a disabled state (finish) of the external power supply function. The disabled state display icon IC4a is turned on (or blinked), when the external power supply function is in a disabled state (particularly, see FIG. 8), and is turned off, when the external power supply function is in an enabled state (particularly, see FIG. 7). Note that, in the present embodiment, the external power supply function is in a disabled state at timing when the work vehicle 1 is activated. However, the external power supply function may be in an enabled state at timing when the work vehicle 1 is activated.

[6. Details of External Power Supply Function]

[0095] Details of the external power supply function are described with reference to FIGS. 9A and 9B. FIGS. 9A and 9B are flowcharts illustrating a flow from start to finish of the external power supply function. The flowchart illustrated in FIG. 9A, and the flowchart illustrated in FIG. 9B are connected by a connecting element A and a connecting element B. The flowchart illustrated in FIG. 9A is started, for example, at timing when the work vehicle 1 is activated. As described above, the external power supply function is in a disabled state at the timing when the work vehicle 1 is

activated. Therefore, in step **S0** (see FIG. **9A**), it is assumed that the external power supply function is in a disabled state (not started).

[0096] As illustrated in FIG. **9A**, in step **S1**, the external power supply control unit **113a4** (see FIG. **4**) determines whether an instruction to start the external power supply function is present. In the present embodiment, the start instruction is achieved by allowing an operator to operate (touch) the enabled state setting icon **IC3** on the state setting screen **113n2** (see FIGS. **7** and **8**). In a case where the start instruction is present (Yes in step **S1**), the processing proceeds to next step **S2**. In a case where the start instruction is not present (No in step **S1**), the external power supply control unit **113a4** continues determination on the start instruction.

[0097] In step **S2**, the external power supply control unit **113a4** determines whether the first external power supply mode is selected, or the second external power supply mode is selected in the external power supply function. In the present embodiment, the selection is achieved by allowing an operator to operate one of the first external power supply mode selection icon **IC1**, and the second external power supply mode selection icon **IC2** on the mode selection screen **113n1** (see FIGS. **5** and **6**). In a case where the first external power supply mode is selected, the processing proceeds to step **S10**, and the first external power supply mode is started as the external power supply function. In a case where the second external power supply mode is selected, the processing proceeds to step **S20**, and the second external power supply mode is started as the external power supply function.

[0098] As illustrated in FIG. **9B**, in step **S11** (after the first external power supply mode is started), the external power supply control unit **113a4** sets the traveling machine body **11** to a state in which external power supply is possible. More specifically, the external power supply control unit **113a4** controls the second inverter **113f**, and supplies alternate-current voltage (e.g., 100 V) to the external power supply port **113g** (see FIG. **3**). As described above, the alternate-current voltage is converted into direct-current voltage (e.g., 350 V), which is supplied from the battery **111d** to the second inverter **113f** by the second inverter **113f**, and the direct-current voltage is generated. Thus, the traveling machine body **11** is brought to a state in which external power supply is possible. Therefore, for example, in this state, connecting the connector of the power supply cable to which the equipment (e.g., an electric work machine or the like) is connected to the external power supply port **113g** enables to supply electric power from the battery **111d** to the equipment. That is, in the first external power supply mode, electric power can be supplied from the battery **111d** to the outside of the traveling machine body **11** (external power supply is possible). When the traveling machine body **11** is set to the state in which external power supply is possible, the processing proceeds to next step **S12**.

[0099] Note that, in the present embodiment, a condition for starting setting on the traveling machine body **11** to a state in which external power supply is possible by the external power supply control unit **113a4** includes presence of a setting switching instruction. In the present embodiment, the setting switching instruction is achieved by operating a switching operation unit (not illustrated) provided in the traveling machine body **11**. That is, when the switching operation unit is operated in a state where the first external power supply mode is started, the traveling machine body **11** is switched between a state in which external power supply is possible, and a state in which external power supply is not possible. The switching operation unit is, for example, a seesaw switch. However, the present invention is not limited to the above configuration, and, for example, a configuration in which the setting switching instruction is omitted may be adopted. That is, it may be configured in such a way that, when the first external power supply mode is started, the external power supply control unit **113a4** sets the traveling machine body **11** to a state in which external power supply is possible (even if a setting switching instruction is not present).

[0100] In step **S12**, power saving control is started. The power saving control is described later. When the power saving control is started, the processing proceeds to next step **S13**.

[0101] In step **S13**, movement prohibition control is started by the traveling control unit **113a2** and

the work machine control unit **113a3** (see FIG. 4 for both). More specifically, in the movement prohibition control, the traveling control unit **113a2** outputs, to the first inverter **113c**, a command of setting the rotational speed of the traveling actuator **111b1** of the traveling drive device **111b** to zero. Further, in the movement prohibition control, the work machine control unit **113a3** controls at least one of the hydraulic pump **114a**, and the direction switching valve **114b** in such a way as to stop driving of the lifting actuator **114c** included in the hydraulic equipment **114**. Further, in the movement prohibition control, the work machine control unit **113a3** outputs, to the first inverter **113c**, a command of setting the rotational speed of the drive actuator **111c1** of the work machine drive device **111c** to zero. Therefore, when the movement prohibition control is started, the traveling machine body **11** is disabled to move. That is, in the first external power supply mode, movement of the traveling machine body **11** is prohibited. Further, movement of the traveling machine body **11** to be prohibited in the first external power supply mode includes traveling of the traveling machine body **11**. When the movement prohibition control is started, the processing proceeds to next step **S14**.

[0102] Note that, the movement prohibition control is not limited to the above configuration. For example, the movement prohibition control may include at least one of control of the lifting actuator **114c**, and control of the drive actuator **111c1** by the work machine control unit **113a3**. That is, movement of the traveling machine body **11** to be prohibited in the first external power supply mode includes at least one of driving of the lifting actuator **114c**, and driving of the drive actuator **111c1**.

[0103] In step **S14**, the traveling control unit **113a2** and the work machine control unit **113a3** determine whether a movement instruction for the traveling machine body **11** is present. In the present embodiment, the movement instruction is achieved by allowing an operator to operate the remote control device **200** (see FIG. 2). For example, allowing an operator to tilt the first operation lever **204a** in the **F1** direction or in the **B1** direction achieves a traveling instruction for the traveling machine body **11**. Further, allowing the operator to tilt the second operation lever **204b** in the **F1** direction or in the **B1** direction achieves a lifting instruction for the work machine **12**, that is, a driving instruction for the lifting actuator **114c**. Further, allowing the operator to operate the second operation switch **205b** achieves a driving instruction for the work machine **12**, that is, a driving instruction for the drive actuator **111c1**.

[0104] Note that, the movement instruction is not limited to one of the traveling instruction for the traveling machine body **11**, the driving instruction for the lifting actuator **114c**, and the driving instruction for the drive actuator **111c1**. For example, the movement instruction may be a combination of two or more of the traveling instruction for the traveling machine body **11**, the driving instruction for the lifting actuator **114c**, and the driving instruction for the drive actuator **111c1**. That is, for example, a case where an operator tilts the second operation lever **204b** in the **F1** direction or in the **B1** direction, while tilting the first operation lever **204a** in the **F1** direction or in the **B1** direction is also included.

[0105] In a case where the movement instruction is present (Yes in step **S14**), the processing proceeds to next step **S15**. In a case where the movement instruction is not present (No in step **S14**), the traveling control unit **113a2** and the work machine control unit **113a3** continue determination on the movement instruction.

[0106] In step **S15**, the traveling machine body **11** is disabled to move due to the movement prohibition control. More specifically, details are as follows. FIG. 10 is an explanatory diagram illustrating an example of movement of the traveling machine body **11** in a case where a movement instruction (a forward movement instruction) is present in the first external power supply mode. For example, when an operator tilts the first operation lever **204a** in the **F1** direction, a traveling instruction (a forward movement instruction) for the traveling machine body **11** is output. However, contrary to the traveling instruction, the traveling control unit **113a2** outputs, to the first inverter **113c**, a command of setting the rotational speed of the traveling actuator **111b1** to zero. Thus, the

rotational speed of the traveling actuator **111b1** becomes zero, and the traveling machine body **11** is maintained in a stopped state (does not travel).

[0107] Note that, even if a driving instruction for the lifting actuator **114c** (a lifting instruction for the work machine **12**) is present, the work machine control unit **113a3** controls at least one of the hydraulic pump **114a**, and the direction switching valve **114b** in such a way that driving of the lifting actuator **114c** is stopped, despite of the driving instruction. Thus, the lifting actuator **114c** stops driving, and the traveling machine body **11** is maintained in a stopped state (the work machine **12** is not raised or lowered).

[0108] Further, even if a driving instruction for the drive actuator **111c1** (a driving instruction for the work machine **12**) is present, the work machine control unit **113a3** outputs, to the first inverter **113c**, a command of setting the rotational speed of the drive actuator **111c1** to zero, despite of the driving instruction. Thus, the rotational speed of the drive actuator **111c1** becomes zero, and the traveling machine body **11** is maintained in a stopped state (the work machine **12** is not driven). Therefore, in the first external power supply mode, even if the traveling machine body **11** is operated by using the remote control device **200**, the traveling machine body **11** is maintained in a stopped state. That is, in the first external power supply mode, an operation of the traveling machine body **11** by the remote control device **200** is prohibited.

[0109] Note that, the movement instruction (particularly, the traveling instruction) may be achieved by automatic traveling control. That is, for example, a traveling instruction for the traveling machine body **11** may be an instruction to be automatically (autonomously) generated by the traveling control unit **113a2** in such a way that the traveling machine body **11** travels along a predetermined route. However, even in this case, the traveling machine body **11** is maintained in a stopped state by the movement prohibition control.

[0110] As illustrated in FIG. 9B, in step **S16**, the external power supply control unit **113a4** determines whether a finishing instruction of the external power supply function is present. In the present embodiment, the finishing instruction is achieved by allowing an operator to operate (touch) the disabled state setting icon **IC4** on the state setting screen **113n2** (see FIGS. 7 and 8). In a case where the finishing instruction is not present (No in step **S16**), the processing is returned to step **S14**. In a case where the finishing instruction is present (Yes in step **S16**), the first external power supply mode as the external power supply function is finished (released) (step **S17**).

[0111] Finishing of the first external power supply mode includes setting on the traveling machine body **11** to a state in which external power supply is not possible by the external power supply control unit **113a4**, finishing of the power saving control, and finishing of the movement prohibition control. Setting on the traveling machine body **11** to a state in which external power supply is not possible by the external power supply control unit **113a4** is achieved by causing the external power supply control unit **113a4** to control the second inverter **113f**, and cutting off supply of alternate-current voltage to the external power supply port **113g**. Therefore, for example, in this state, even if the connector of the power supply cable to which the equipment (e.g., an electric work machine or the like) is connected is connected to the external power supply port **113g**, electric power from the battery **111d** is not supplied to the equipment.

[0112] Therefore, in the present embodiment, the first external power supply mode is released by an operation of the disabled state setting icon **IC4**. That is, in the present embodiment, the disabled state setting icon **IC4** also has a function as a release operation unit **115** that performs setting on releasing of the first external power supply mode (see FIGS. 7 and 8). Further, as described above, the disabled state setting icon **IC4** as the release operation unit **115** is provided on the operation unit **113n** (particularly, on the state setting screen **113n2**) disposed in the work vehicle **1**. That is, the work vehicle **1** includes the release operation unit **115**.

[0113] Note that, the release operation unit **115** is not limited to the above configuration. For example, the release operation unit **115** may have a configuration in which a dedicate icon for setting on releasing of the first external power supply mode is provided, or may have a

configuration in which an input of a password is requested. Further, the release operation unit **115** may be provided independently of the operation unit **113n**, or may be constituted of a switch, a dial, a lever, or the like.

[0114] In step **S21** (after the second external power supply mode is started), similarly to step **S11**, the external power supply control unit **113a4** sets the traveling machine body **11** to a state in which external power supply is possible. That is, similarly to the first external power supply mode, also in the second external power supply mode, electric power can be supplied from the battery **111d** to the outside of the traveling machine body **11** (external power supply is possible). Note that, since the processing of step **S21** is similar to the processing of step **S11**, detailed description thereof is omitted. When the traveling machine body **11** is set to a state in which external power supply is possible, the processing proceeds to next step **S22**.

[0115] Note that, for example, in the second external power supply mode, connecting the connector of the power supply cable to the external power supply port **113g** enables to supply electric power to the equipment to which the power supply cable is connected. However, at this occasion, even if the connector is detached from the external power supply port **113g**, the second external power supply mode is continued (not released).

[0116] Further, in the present embodiment, as described above (in step **S11**), the condition for starting setting on the traveling machine body **11** to a state in which external power supply is possible by the external power supply control unit **113a4** includes presence of a setting switching instruction. That is, when the switching operation unit is operated in a state where the second external power supply mode is started, the traveling machine body **11** is switched between a state in which external power supply is possible, and a state in which external power supply is not possible. However, the present invention is not limited to the above configuration, and, for example, a configuration in which a setting switching instruction is omitted may be adopted. That is, as far as the second external power supply mode is started, the external power supply control unit **113a4** may set the traveling machine body **11** to a state in which external power supply is possible (even if a setting switching instruction is not present).

[0117] In step **S22**, similarly to step **S14**, the traveling control unit **113a2** and the work machine control unit **113a3** determine whether a movement instruction for the traveling machine body **11** is present. Note that, since the processing of step **S22** is similar to the processing of step **S14**, detailed description thereof is omitted. In a case where the movement instruction is present (Yes in step **S22**), the processing proceeds to next step **S23**. In a case where the movement instruction is not present (No in step **S22**), the traveling control unit **113a2** and the work machine control unit **113a3** continue determination on the movement instruction.

[0118] In step **S23**, at least one of the traveling control unit **113a2**, and the work machine control unit **113a3** controls the traveling machine body **11** in response to a movement instruction for the traveling machine body **11**, and causes the traveling machine body **11** to move. That is, movement of the traveling machine body **11** is permitted in the second external power supply mode. More specifically, details are as follows. FIG. **11** is an explanatory diagram illustrating an example of movement of the traveling machine body **11** in a case where a movement instruction (a forward movement instruction) is present. For example, by allowing an operator to tilt the first operation lever **204a** in the **F1** direction, a traveling instruction (a forward movement instruction) for the traveling machine body **11** is output. Then, the traveling control unit **113a2** controls the traveling actuator **111b1** via the first inverter **113c**, based on the traveling instruction. That is, for example, the traveling control unit **113a2** outputs, to the first inverter **113c**, a command of setting the rotational speed of the traveling actuator **111b1** to a rotational speed associated with the traveling instruction. Thus, the traveling machine body **11** travels based on the traveling instruction.

[0119] Note that, in a case where a driving instruction for the lifting actuator **114c** (a lifting instruction for the work machine **12**) is present, the work machine control unit **113a3** controls the hydraulic pump **114a** and the direction switching valve **114b**, based on the driving instruction.

More specifically, the work machine control unit **113a3** drives the hydraulic pump **114a**, and switches the direction switching valve **114b** in response to the driving instruction. Thus, the lifting actuator **114c** is driven based on the driving instruction, and the work machine **12** is raised or lowered.

[0120] Further, in a case where a driving instruction for the drive actuator **111cl** (a driving instruction for the work machine **12**) is present, the work machine control unit **113a3** controls the drive actuator **111c1** via the first inverter **113c**, based on the driving instruction. More specifically, the work machine control unit **113a3** outputs, to the first inverter **113c**, a command of setting the rotational speed of the drive actuator **111c1** to a predetermined rotational speed (e.g., a rated rotational speed). Thus, the drive actuator **111cl** is driven based on the driving instruction, and the work machine **12** is driven.

[0121] Note that, the movement instruction (particularly, the traveling instruction) may be achieved by automatic traveling control. That is, for example, a traveling instruction for the traveling machine body **11** may be an instruction to be automatically (autonomously) generated by the traveling control unit **113a2** in such a way that the traveling machine body **11** travels along a predetermined route. In this case, the traveling machine body **11** performs automatic traveling.

[0122] As illustrated in FIG. **9B**, in step **S24**, similarly to step **S16**, the external power supply control unit **113a4** determines whether a finishing instruction of the external power supply function is present. Note that, since the processing of step **S24** is similar to the processing of step **S16**, detailed description thereof is omitted. In a case where the finishing instruction is not present (No in step **S24**), the processing is returned to step **S22**. In a case where the finishing instruction is present (Yes in step **S24**), the second external power supply mode as the external power supply function is finished (step **S25**). Finishing of the second external power supply mode includes setting on the traveling machine body **11** to a state in which external power supply is not possible by the external power supply control unit **113a4** (see step **S17**).

[0123] According to the above configuration, in the work vehicle **1**, when the first external power supply mode is started, the traveling machine body **11** is brought to a state in which electric power of the battery **111d** can be supplied to the outside of the traveling machine body **11**, and movement of the traveling machine body **11** is prohibited. Therefore, in a state where electric power of the battery **111d** can be supplied to the outside of the traveling machine body **11** (a state in which external power supply is possible), for example, movement of the traveling machine body **11** due to an erroneous operation can be prevented. Thus, when the traveling machine body **11** is used as a power source, it is possible to prevent a user of the power source (the traveling machine body **11**) located near the traveling machine body **11** from being damaged by movement of the traveling machine body **11**. That is, it is possible to secure safety of the traveling machine body **11** in a state where external power supply is possible.

[0124] Further, in the work vehicle **1**, when the second external power supply mode is started, since movement of the traveling machine body **11** is permitted, the traveling machine body **11** can be moved, while electric power of the battery **111d** is supplied to the outside of the traveling machine body **11** (external power supply is performed). Thus, for example, an electric work machine (a pesticide spraying machine or the like) can be electrically connected to the traveling machine body **11**, and the work machine can perform work by being driven with electric power of the battery **111d**. Alternatively, electrically connecting a chargeable mobile terminal (a personal computer or the like) different from the traveling machine body **11** to the traveling machine body **11** enables to charge the mobile terminal, while performing work by using the traveling machine body **11**. That is, it is possible to secure convenience of the traveling machine body **11** in a state where external power supply is possible. Thus, it is possible to secure safety and convenience of the traveling machine body **11** in a state where external power supply is possible.

[0125] In the first external power supply mode, prohibiting traveling of the traveling machine body **11** among movement of the traveling machine body **11** enables to avoid traveling of the traveling

machine body **11** in a state where external power supply is possible, and contact with a user (of the power source) located near the traveling machine body **11**. In this viewpoint, as described in the present embodiment, it is desirable that movement of the traveling machine body **11** to be prohibited in the first external power supply mode includes traveling of the traveling machine body **11**.

[0126] In a configuration in which the traveling machine body **11** includes the lifting actuator **114c** that raises and lowers the work machine **12**, prohibiting driving of the lifting actuator **114c** in the first external power supply mode makes it impossible for the work machine **12** to rise and fall (the work machine **12** is not raised or lowered). Thus, it is possible to avoid that the work machine **12** is raised and lowered with the traveling machine body **11** being in a state where external power supply is possible, and contact of the work machine **12** with a user (of the power source) located near the work machine **12**. Further, in a configuration in which the traveling machine body **11** includes the drive actuator **111c1** that drives the work machine **12**, prohibiting driving of the drive actuator **111c1** in the first external power supply mode makes it impossible for the work machine **12** to drive (the work machine **12** is not driven). Thus, it is possible to avoid that the work machine **12** is driven with the traveling machine body **11** being in a state where external power supply is possible, and contact of the work machine **12** with a user located near the work machine **12**. For example, as illustrated in FIG. **1**, in a case where the work machine **12** is a tiller, contact of a claw (not illustrated) of the tiller in pivotal movement with a user located near the tiller is avoided. From a viewpoint as described above, as described in the present embodiment, it is desirable that movement of the traveling machine body **11** to be prohibited in the first external power supply mode includes at least one of driving of the lifting actuator **114c**, and driving of the drive actuator **111c1**.

[0127] In a configuration in which the work vehicle **1** includes the remote control device **200** that operates the traveling machine body **11**, it is desirable to prevent the traveling machine body **11** in a state where external power supply is possible from moving due to an erroneous operation of the remote control device **200**. In this viewpoint, as described in the present embodiment, it is desirable that an operation of the traveling machine body **11** by the remote control device **200** is prohibited in the first external power supply mode.

[0128] As far as the first external power supply mode is released via a manual release operation, it is possible to avoid that the first external power supply mode is automatically released, and the traveling machine body **11** starts to move and damages a user, despite that the user (of the power source) is located near the traveling machine body **11**. In this viewpoint, as described in the present embodiment, it is desirable that the work vehicle **1** includes the release operation unit **115** that performs setting on releasing of the first external power supply mode.

[0129] Herein, the power saving control (see step **S12** in FIG. **9B**) is described with reference to FIG. **12**. FIG. **12** is an explanatory diagram illustrating the power saving control. Note that, in FIG. **12**, a hatched portion indicates target equipment for the power saving control.

[0130] The power saving control is provided to suppress electric power consumption of the traveling machine body **11**. More specifically, when the power saving control is started, electric power consumption of the plurality of pieces of electric equipment **113** other than those required for the external power supply function (including the movement prohibition control) is suppressed. That is, for example, power supply of the positioning communication unit **113k** is turned off, or the operation unit **113n** is brought to a sleep state. Note that, the sleep state means a state in which a screen is not displayed, despite that power supply is in an on-state.

[0131] As described above, the positioning communication unit **113k** is used for acquiring position information (e.g., a latitude and a longitude) of the vehicle body **10** (particularly, during automatic traveling), and is not used for the external power supply function. Therefore, in the external power supply function, the positioning communication unit **113k** is not necessary. In the present embodiment, switching of power supply of the positioning communication unit **113k** to an off-state

is achieved by outputting a switching command to the off-state from the control device **113a** to the positioning communication unit **113k**. Note that, the above switching may be achieved by cutting off supply of electric power to the positioning communication unit **113k**. Thus, electric power consumption of the traveling machine body **11** is reduced by the electric power consumption of the positioning communication unit **113k**.

[0132] Since the operation unit **113n** is constituted of a touch panel as described above, the operation unit **113n** consumes electric power for screen display and the like. Further, the operation unit **113n** enables various settings on the traveling machine body **11**. However, when the first external power supply mode is started, the traveling machine body **11** is used as a power source. Therefore, there is no needs to operate the operation unit **113n** other than an operation of finishing the first external power supply mode, and the operation unit **113n** is not necessary. In the present embodiment, switching of the operation unit **113n** to a sleep state is achieved by detecting that a non-operation state continues for a predetermined time (e.g., one minute) in the operation unit **113n**. Note that, when the operation unit **113n** is operated (touched) in the sleep state, a screen is displayed. Thus, electric power consumption of the traveling machine body **11** is reduced, as compared with a case where a screen of the operation unit **113n** is continuously displayed. Note that, instead of setting the operation unit **113n** to a sleep state, power supply of the operation unit **113n** may be turned off. However, also in this case, it is preferable that, when the operation unit **113n** in a power-off state is operated, power supply of the operation unit **113n** is switched to an on-state, and a screen is displayed.

[0133] From a viewpoint that electric power for external power supply increases among electric power stored in the battery **111d**, as illustrated in FIGS. **9B** and **12**, it is desirable that electric power consumption of the traveling machine body **11** is suppressed in the first external power supply mode.

[0134] Note that, the electric equipment **113** in which electric power consumption is suppressed by the power saving control is not limited to the positioning communication unit **113k**, and the operation unit **113n**. For example, in a case where the control device **113a** is configured of a plurality of pieces of hardware, electric power consumption of the plurality of pieces of hardware other than those for use in the external power supply function may be suppressed. Thus, electric power consumption of the traveling machine body **11** is reduced by the electric power consumption of the hardware (hardware in which electric power consumption is suppressed).

[7. Supplementary Description]

[0135] In the present embodiment, a configuration in which movement prohibition control is started (performed), in a case where the traveling machine body **11** is in a state where external power supply is possible has been described. However, timing when the movement prohibition control is started is not limited to the above. For example, the movement prohibition control may not be started when the traveling machine body **11** is in a state where external power supply is possible, but may be started at timing when electric power is supplied from the battery **111d** to the outside of the traveling machine body **11**.

[8. Supplementary Notes]

[0136] The work vehicle **1** described in the present embodiment can also be expressed as a work vehicle described in the following supplementary notes.

[0137] A work vehicle of supplementary note (1) is [0138] a work vehicle provided with a traveling machine body in which a battery is disposed, the work vehicle including [0139] a first external power supply mode and a second external power supply mode in which electric power can be supplied from the battery to the outside of the traveling machine body, wherein [0140] in the first external power supply mode, movement of the traveling machine body is prohibited, and [0141] in the second external power supply mode, movement of the traveling machine body is permitted.

[0142] A work vehicle of supplementary note (2) is the work vehicle according to supplementary note (1), wherein [0143] movement of the traveling machine body includes traveling of the

traveling machine body.

[0144] A work vehicle of supplementary note (3) is the work vehicle according to supplementary note (1) or (2), further including [0145] a work machine to be connected to the traveling machine body, wherein [0146] the traveling machine body includes [0147] a lifting actuator that raises and lowers the work machine, and [0148] a drive actuator that drives the work machine, and [0149] movement of the traveling machine body includes at least one of driving of the lifting actuator and driving of the drive actuator.

[0150] A work vehicle of supplementary note (4) is the work vehicle according to any one of supplementary notes (1) to (3), further including [0151] a remote control device that operates the traveling machine body, wherein [0152] in the first external power supply mode, an operation of the traveling machine body by the remote control device is prohibited.

[0153] A work vehicle of supplementary note (5) is the work vehicle according to any one of supplementary notes (1) to (4), wherein [0154] in the first external power supply mode, electric power consumption of the traveling machine body is suppressed.

[0155] A work vehicle of supplementary note (6) is the work vehicle according to any one of supplementary notes (1) to (5), further including [0156] a release operation unit that performs setting on releasing of the first external power supply mode.

[0157] The embodiment of the present invention has been described above, but the scope of the present invention is not limited thereto, and may be expanded or modified without departing from the gist of the invention.

INDUSTRIAL APPLICABILITY

[0158] The present invention is applicable, for example, to a work vehicle such as an agricultural machine and a construction machine.

REFERENCE SIGNS LIST

[0159] **1** Work vehicle [0160] **11** Traveling machine body [0161] **12** Work machine [0162] **111c1** Drive actuator [0163] **111d** Battery [0164] **114c** Lifting actuator [0165] **115** Release operation unit [0166] **200** Remote control device

Claims

1. A work vehicle provided with a traveling machine body in which a battery is disposed, the work vehicle comprising: a first external power supply mode and a second external power supply mode in which electric power can be supplied from the battery to the outside of the traveling machine body, wherein in the first external power supply mode, movement of the traveling machine body is prohibited, and in the second external power supply mode, movement of the traveling machine body is permitted.
2. The work vehicle according to claim 1, wherein movement of the traveling machine body includes traveling of the traveling machine body.
3. The work vehicle according to claim 1, further comprising: a work machine to be connected to the traveling machine body, wherein the traveling machine body includes: a lifting actuator that raises and lowers the work machine, and a drive actuator that drives the work machine, wherein movement of the traveling machine body includes at least one of driving of the lifting actuator and driving of the drive actuator.
4. The work vehicle according to claim 1, further comprising: a remote control device that operates the traveling machine body, wherein in the first external power supply mode, an operation of the traveling machine body by the remote control device is prohibited.
5. The work vehicle according to claim 1, wherein: in the first external power supply mode, electric power consumption of the traveling machine body is suppressed.

6. The work vehicle according to claim 1, further comprising: a release operation unit that performs setting on releasing of the first external power supply mode.
